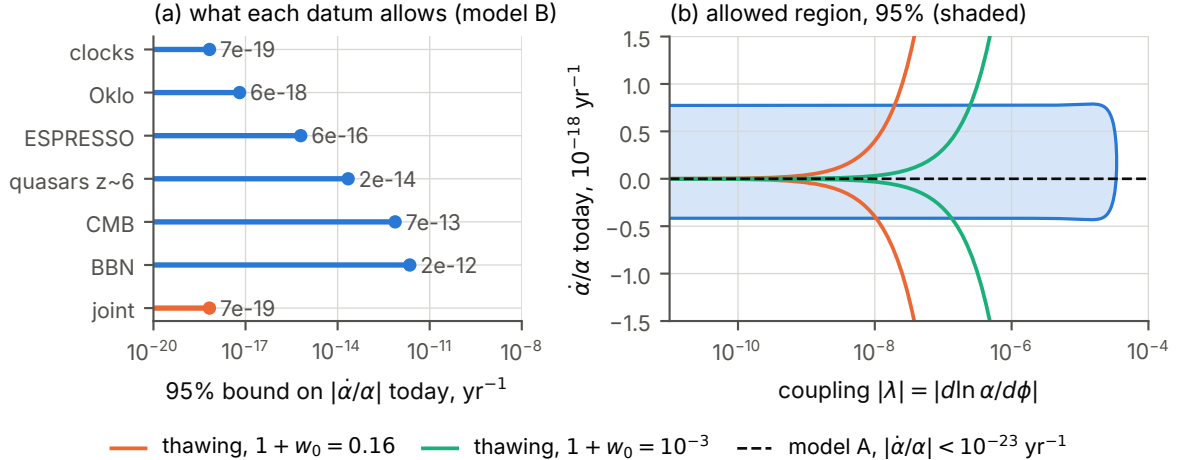
(a) what each datum allows (model B)
clocks
Oklo
ESPRESSO
quasars z~6
CMB
BBN
joint
7e-19
6e-18
6e-16
2e-14
7e-13
2e-12
7e-19
$10^{-20}$
$10^{-17}$
$10^{-14}$
$10^{-11}$
$10^{-8}$
95% bound on $|\dot{\alpha}/\alpha|$ today, $\mathrm{yr}^{-1}$
(b) allowed region, 95% (shaded)
$\dot{\alpha}/\alpha$ today, $10^{-18}\ \mathrm{yr}^{-1}$
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
$10^{-10}$
$10^{-8}$
$10^{-6}$
$10^{-4}$
coupling $|\lambda| = |d\ln\alpha/d\phi|$
thawing, $1 + w_0 = 0.16$
thawing, $1 + w_0 = 10^{-3}$
model A, $|\dot{\alpha}/\alpha| < 10^{-23}\ \mathrm{yr}^{-1}$